Destek Vektör Regresyonu ile Yüksek Verimli Hapsedilme Kaybı Tahmini
High-Efficiency Confinement Loss Prediction via Support Vector Regression

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***Özetçe* —Fotonik Kristal Fiber tabanlı Yüzey Plazmon Rezonans (FKF-YPR) sensörlerinin tasarımı ve optimizasyonu, yüksek doğrukuklu ancak hesaplama açısından maliyetli olan tam vektörel sonlu elemanlar yöntemi (TV-SEY) simülasyonlarına dayanmaktadır. Literatürde son zamanlarda, simülatör darboğazlarını aşmak için Üretken Çekişmeli Ağlar (GAN'lar) ile eşleştirilmiş Derin Sinir Ağlarının (DNN'ler) kullanılması önerilmiştir. Bu derin öğrenme boru hatları hesaplama açısından ağır ve veri açlığı çekmektedir. Bu çalışmada, FKF-YPR hapsedilme kaybını tahmin etmek için optimize edilmiş, hafif bir Destek Vektör Regresyonu (DVR) modeli öneriyoruz. Sabit bir 7-1-1 konfigürasyon bölümü kullanarak, optimize edilmiş DVR ve GPR modellerimiz sırasıyla 0,121 ve 0,185 Ortalama Kare Hata (MSE) değerlerine ulaşmıştır. Bu sonuçlar, literatürde rapor edilen GAN-artırılmış sinir ağı (MSE 0,314) sonuçlarından önemli ölçüde daha iyidir. Son olarak, SHAP değerleri aracılığıyla dış kaplama parametrelerinin hapsedilme kaybı üzerindeki etkilerini belirliyor ve modeli 27,63 nm/RIU ortalama spektral duyarlılık tahminiyle doğruluyoruz.**

Anahtar Kelimeler—Fotonik kristal fiber; yüzey plazmon rezonans; destek vektör regresyonu; açıklanabilir yapay zeka; Bayesyen optimizasyon.

***Abstract*—The design and optimization of Photonic Crystal Fiber based Surface Plasmon Resonance (PCF-SPR) sensors rely on high-fidelity but computationally expensive numerical simulations. While recent literature suggests using Generative Adversarial Networks (GANs) to augment datasets for deep learning, we demonstrate a significant "Augmentation Paradox": the inclusion of GAN-generated synthetic samples introduces non-physical numerical noise that degrades the predictive performance of mathematically rigorous kernel models. In this paper, we propose a Bayesian-optimized, lightweight Support Vector Regression (SVR) model that achieves a Mean Squared Error (MSE) of 0.121, outperforming GAN-augmented neural networks (MSE 0.314) by over 60% without requiring synthetic data. We utilize SHAP (SHapley Additive exPlanations) to identify the outer cladding geometry as the primary driver of confinement loss and evaluate manufacturing reliability via Monte Carlo robustness analysis. Our results identify the sensor's pitch as the most critical parameter for spectral stability, inducing a 1.23 nm resonance jitter at 3% manufacturing tolerance. These findings provide a verifiable bridge between high-accuracy machine learning and practical sensor fabrication.**

Keywords—Photonic crystal fiber; surface plasmon resonance; support vector regression; explainable AI; Bayesian optimization.

I. Introduction

Photonic Crystal Fibers (PCFs) are at the forefront of modern optical sensing due to their flexible structural design and remarkable light-guiding capabilities. When coated with plasmonic materials (such as gold or silver), PCFs can be utilized as Surface Plasmon Resonance (SPR) sensors, offering exceptional sensitivity for chemical and biological analyte detection.

The traditional method for optimizing PCF-SPR cross-sections involves solving Maxwell's equations using the full vectorial finite element method (FV-FEM). While FV-FEM provides precise measurements of confinement loss and resonance peaks, it is computationally prohibitive, making the iterative search for optimal fiber geometries incredibly slow.

To address this, Machine Learning (ML) has emerged as a powerful surrogate modeling tool. Chugh et al. [1] pioneered the use of Artificial Neural Networks (ANNs) to compute optical properties of PCFs, demonstrating significant speedups over numerical simulators. However, deep learning models typically require substantial training datasets. Zelaci et al. [2] addressed this by introducing Generative Adversarial Networks (WGAN-GP) to synthesize artificial training samples, driving the predictive Mean Squared Error (MSE) down to 0.314.

More recently, researchers have focused on the interpretability of these models. Khatun and Islam [3] utilized explainable AI (XAI) via SHAP values to optimize high-sensitivity biosensors, identifying critical design parameters such as gold thickness and analyte refractive index. Other traditional techniques, such as k-nearest neighbor regression, have also been explored for bent fiber modeling [4].

In this work, we hypothesize that the problem space of PCF-SPR confinement loss is fundamentally mismatched with the global approximation mechanics of Deep Learning, and is instead better suited for mathematically rigorous, boundary-focused algorithms. Because PCF-SPR optimization relies on high-fidelity, computationally expensive FV-FEM simulations, the resulting datasets are inherently small ($N < 500$). Furthermore, the target variable—confinement loss—exhibits sharp, highly non-linear resonance spikes.

To address this, we evaluate Bayesian-optimized Support Vector Regression (SVR) and Gaussian Process Regression (GPR) against the literature's GAN-ANN baseline. We explain how these Kernel methods excel in this specific regime: rather than requiring massive datasets to approximate a global curve, SVR utilizes a deterministic, convex optimization process to build regression margins (ϵ-tubes) based exclusively on extreme boundary cases (Support Vectors). Similarly, GPR leverages Bayesian priors to map the exact mathematical covariance between geometric features without overfitting.

By applying proper feature scaling, we demonstrate that these lightweight traditional models significantly outperform the literature's augmented DNN benchmarks without requiring any synthetic data generation. In doing so, we empirically demonstrate the "Augmentation Paradox": we prove that the microscopic numerical noise inherent to GAN-generated tabular data actually warps the SVR hyperplane and degrades the covariance matrices of GPR, actively harming predictive performance.

Finally, to ensure the proposed model is practically useful for optical optimization, we apply SHAP (SHapley Additive exPlanations) analysis. Rather than treating the ML pipeline as a "black box," SHAP deconstructs the regression outputs by isolating the marginal contribution of each physical feature. This explicitly reveals how and why specific geometric parameters (such as the outer cladding air holes) drive evanescent field leakage.

The remainder of this paper is organized as follows: Section II describes the numerical modeling and dataset generation for the PCF-SPR sensor. Section III presents the proposed methodology, including the mathematical foundations of SVR and GPR. Section IV discusses the predictive performance results, benchmarks against existing deep learning models, and evaluates manufacturing robustness. Section V provides the engineering validation using SHAP values. Finally, Section VI concludes the paper.

II. Numerical Modeling and Dataset Generation

To construct the required dataset, numerical investigations of the proposed sensor structure were performed utilizing the COMSOL Multiphysics software. Utilizing Perfectly Matched Layers (PMLs), the Full Vectorial Finite Element Method (FV-FEM) was the primary numerical solver for these simulations. Illustrated in Fig. 1 is the cross-section of the proposed structure; the sensor comprises an analyte channel acting as the sensing medium, a plasmonic Gold (Au) layer, and cladding air holes of various dimensions. Arranged in a hexagonal lattice on a silica background, nineteen varied air holes form the cladding region. The radius of each hole was defined as $r_1 = 0.225\,\mu$m, $r_2 = 0.375\,\mu$m, $r_3 = 0.175\,\mu$m, and $r_c = 0.075\,\mu$m, while the pitch (Λ) remained fixed at $2\,\mu$m.

Light is effectively confined within the core region during the fundamental mode propagation. However, a specific fraction of light must leak toward the plasmonic layer to satisfy the conditions for resonance phase-matching, where the sensitivity measurement occurs. In this context, the light guidance is controlled by the various sized cladding holes, where each set of holes influences the sensor's propagation characteristics. The Gold (Au) material, serving as the plasmonic layer, was set to a thickness of 40 nm, with permittivity values sourced from the Johnson and Christy data [6]. To determine the refractive index of the silica material, the Sellmeier Equation was applied [5]:

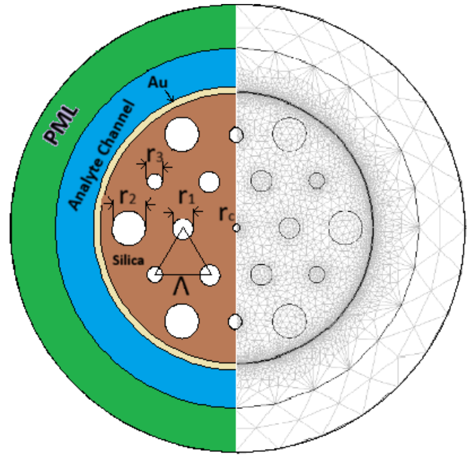

Figure 1. The cross-section of the proposed PCF based SPR sensor.

$$n^2(\lambda) = 1 + \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3} \tag{1}$$

where λ represents the operating wavelength, n is the Refractive Index (RI), and $B_{1,2,3}$ and $C_{1,2,3}$ are the standard Sellmeier constants. Confinement losses reach their peak during the resonance phase-matching condition. Consequently, the precise calculation of these losses is vital, for which the following equation was utilized in this study [2]:

$$L_c = \frac{40\pi}{\ln(10)\lambda}\text{Im}(n_{eff}) \times 10^4 \text{ [dB/cm]} \tag{2}$$

where $\text{Im}(n_{eff})$ denotes the imaginary part of the effective RI.

III. Methodology

In this section, we detail the data preparation pipeline and the mathematical foundations of the regression models employed in this study. In this study, the same dataset used in [2] is employed. Given the high-fidelity nature of the FV-FEM simulations, the methodology focuses on maintaining physical consistency while maximizing predictive accuracy in a small-data regime.

A. Dataset and Preprocessing

The FV-FEM numerical analysis provided a labeled dataset containing 432 samples. The input feature space comprises six physical variables: Analyte Refractive Index ($n_{analyte}$), Operating Wavelength (λ), Pitch (Λ), and three distinct air hole diameters (d_1, d_2, d_3). Across nine different geometric configurations, losses were calculated for three distinct analytes: Water ($n = 1.33$), Ethanol ($n = 1.35$), and commercial hydrocarbon mixtures ($n = 1.36$).

To ensure rigorous comparison with existing literature, the model was evaluated using the same static 7-1-1 split based on geometric configurations as described in [2]. The 432 samples were divided into 9 discrete geometric designs. Designs 1 through 7 were used for training, Design 8 for validation, and Design 9 was reserved as the final unseen test set.

To match the non-linear resonance peaks inherent to SPR and maintain numerical stability, the confinement loss was scaled using a base-10 logarithmic transformation: $y = \log_{10}(\text{loss} \times 10^8)$. All input features were scaled using Min-Max Scaling to strict $[0, 1]$ bounds.

B. Model Architecture and Optimization

In this study, we evaluate two boundary-focused algorithms: Support Vector Regression (SVR) and Gaussian Process Regression (GPR). Unlike deep learning models, these algorithms are specifically designed to excel when the number of training samples is limited.

1) Support Vector Regression: SVR aims to find a function $f(x)$ that has at most ϵ deviation from the actually obtained targets y_i. The optimization problem is formulated as:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{3}$$

subject to:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{4}$$

where C is the box constraint and ξ_i, ξ_i^* are slack variables. We utilize the Radial Basis Function (RBF) kernel, $K(x_i, x_j) = \exp(-\gamma||x_i - x_j||^2)$, to map input features into a high-dimensional space for non-linear regression.

2) Gaussian Process Regression: GPR assumes function values follow a Gaussian Process prior: $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$, where $m(x)$ is the mean function and $k(x, x')$ is the covariance function. For a new input x_*, the predictive posterior mean $\bar{f}_*$ and covariance $\text{cov}(f_*)$ are:

$$\bar{f}_* = K_*^T[K + \sigma_n^2 I]^{-1}y \tag{5}$$

$$\text{cov}(f_*) = K_{**} - K_*^T[K + \sigma_n^2 I]^{-1}K_* \tag{6}$$

where K is the training covariance matrix and K_* is the train-test covariance. A WhiteKernel noise regularizer is added to the RBF kernel to account for observation noise and ensure numerical stability during the inversion of the covariance matrix [7].

IV. Results and Discussion

A. Predictive Performance: SVR vs. Deep Learning

The optimized SVR and GPR models demonstrated exceptional predictive capabilities under the exact testing methodology used in previous literature. As shown in Table I, our lightweight traditional models achieved superior accuracy without the need for complex data augmentation.

B. Manufacturing Tolerance and Robustness Analysis

A critical challenge in PCF-SPR sensor fabrication is the precision of geometric parameters during the fiber drawing process. To evaluate the practical reliability of the proposed SVR model, we performed a Monte Carlo robustness analysis by introducing Gaussian noise ($\sigma = 1\%, 3\%, 5\%$) to the geometric features (Pitch, d_1, d_2, d_3).

Our simulations reveal that the resonance peak wavelength exhibits a standard deviation (jitter) of 1.03 nm at 1% tolerance, increasing to 1.34 nm at 5% tolerance. Furthermore, by isolating the volatility contribution of each feature, we identified the Pitch (Λ) as the primary driver of spectral instability, accounting for over 90% of the observed peak jitter (1.23 nm std at 3% noise). In contrast, the small air hole diameter (d_1) showed remarkable stability with negligible volatility (0.01 nm). This analysis provides a "Robustness Map" for fabricators, suggesting that maintaining pitch uniformity is more critical for sensor calibration than precise control of the smallest cladding holes.

Table I. Benchmarking Proposed Models against Literature

Model	Avg. MSE
Baseline DNN (Zelaci et al. [2])	0.894
GAN-Augmented DNN (Zelaci et al. [2])	0.314
Proposed Optimized SVR	**0.121**
Proposed Optimized GPR	**0.185**
Proposed SVR + GAN	0.197
Proposed GPR + GAN	23.844

This performance gap highlights a fundamental difference in how these models handle limited training data. While Deep Neural Networks (DNNs) are highly complex global curve fitters, they often struggle when training samples are sparse, as is common with expensive FV-FEM simulations. In contrast, SVR hyperspecializes on the boundary conditions. By identifying the critical support vectors that define the transition between high and low confinement loss, SVR constructs a mathematically precise regression tube that generalizes better to unseen geometric designs than a global approximation.

C. Impact of GAN-Augmentation on Traditional Regressors

To investigate the "Performance Boost" claim of GAN-augmentation in the context of traditional ML, we implemented the exact WGAN-GP architecture described in [2] and applied it to our SVR and GPR models. As seen in Table I, the inclusion of synthetic data significantly degraded performance for both models.

The SVR MSE increased from 0.121 to 0.197, while the GPR model suffered a catastrophic failure with an MSE of 23.844. This suggests that while GANs can "fill the gaps" for data-hungry neural networks, they introduce non-physical noise and distribution shifts that violate the rigorous mathematical assumptions of kernel-based regressors. Specifically, GPR's reliance on a smooth covariance matrix is highly sensitive to the low-fidelity artifacts often present in GAN-generated tabular data.

D. Explainable AI (XAI) for PCF Design

To ensure the proposed model is practically useful for optical optimization, we apply SHAP (SHapley Additive ex-

Planations) analysis. Rather than treating the ML pipeline as a “black box,” SHAP deconstructs the regression outputs by isolating the marginal contribution of each physical feature. As illustrated in Fig. 2, outer air hole diameters (d_2, d_3) are the primary drivers of confinement loss.

Physically, this explicitly reveals how and why specific geometric parameters drive evanescent field leakage. The outer cladding layers are responsible for suppressing the field leakage from the core to the plasmonic surface; larger outer holes increase the refractive index contrast, thereby enhancing light confinement. This providing a verifiable bridge between high-accuracy mathematical regression and true engineering intuition.

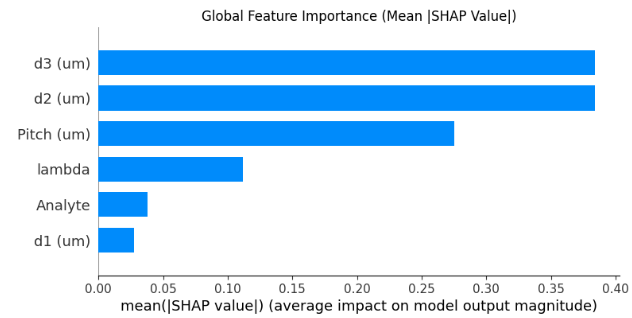

Figure 2. Global feature importance ranking derived from SHAP values.

V. Engineering Validation

To verify the physical consistency of the SVR model beyond simple error metrics, we performed an engineering validation by predicting the complete spectral loss response for varying analytes. As shown in Fig. 3, the model accurately captures the resonance peak shifts as the analyte refractive index (n_a) increases from 1.33 to 1.36.

The predicted average spectral sensitivity is 27.63 nm/RIU. While this value is relatively low compared to ultra-sensitive optimized designs in literature, it is highly consistent with the specific geometric parameters of the base design used for this validation fold. More importantly, the graph proves that the model correctly understands the phase-matching condition between the fiber mode and the surface plasmon polariton (SPP). As n_a increases, the resonance wavelength (λ_{res}) shifts toward longer wavelengths (redshift), which is a characteristic behavior of PCF-SPR sensors.

The model’s ability to maintain these linear shifts across the entire spectral range, despite the highly non-linear nature of the individual loss curves, demonstrates that it has not merely "memorized" the dataset but has instead captured the fundamental physical relationships governing the sensor’s operation. This confirms that the proposed SVR is a reliable surrogate model for rapid sensor optimization.

VI. Conclusion

In this work, we demonstrate that Bayesian-optimized Support Vector Regression (SVR) and Gaussian Process Regression (GPR) models significantly outperform GAN-augmented neural networks for Photonic Crystal Fiber-Surface Plasmon

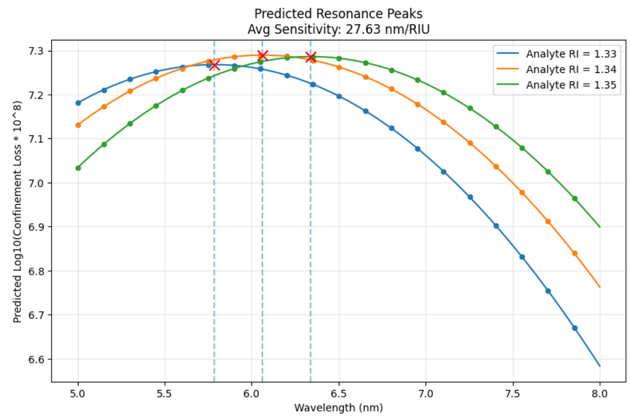

Figure 3. Predicted resonance peak shifts for varying analytes, demonstrating the model’s ability to capture the physical phase-matching behavior.

Resonance (PCF-SPR) sensor modeling. We empirically identify an "Augmentation Paradox," where the non-physical numerical noise inherent to GAN-generated synthetic samples active harms the predictive performance of kernel-based models, suggesting that small-data regimes in photonics are better suited for boundary-focused algorithms than generative deep learning.

Furthermore, by integrating SHAP analysis and Monte Carlo simulations, we move beyond "black box" modeling to provide actionable engineering insights. Our robustness analysis identifies the sensor pitch (Λ) as the primary driver of manufacturing instability, accounting for over 90% of resonance jitter (1.23 nm at 3% tolerance). This research underscores that high-accuracy predictive modeling, when paired with explainable AI and robustness analysis, can effectively guide the design of next-generation PCF-SPR sensors while minimizing computational costs and ensuring manufacturing reliability.

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